Jan 27

- Recall group def: Let G be a set and $*: G \times G \to G$ a binary operation. (implies closure)
  We say G is a group if:
  1) $(a*b)*c = a*(b*c)$ Associativity
  2) $\exists\, e \in G$ s.t. $a*e = e*a = a \quad \forall a \in G$ Identity
  3) $\forall a \in G \ \exists\, a^{-1} \in G$ s.t. $a*a^{-1} = a^{-1}*a = e$ Inverses
     (not necessarily $\frac{1}{a}$)

  Additionally, if $a*b = b*a \ \forall a,b \in G$ we say G is Abelian

- <u>Thm 2.1</u> In a group, there is a <u>unique</u> identity element
  <u>Proof</u> Suppose e and e' both satisfy the identity condition.
  Let $a \in G$. We have $a*e = a*e' = a$. Multiplying by inverses is often the ace in our sleeve
  We group operation on the left by a^{-1}: $a^{-1}*(a*e) = a^{-1}*(a*e')$.
  (group is not necessarily Abelian, so we don't assume commutativity)

  By associativity, we get $(a^{-1}*a)*e = (a^{-1}*a)*e'$. By inverses, $e*e = e*e'$.
  Since e is an identity, $e = e'$. Thus, the identity is unique. $\square$

- We will drop the $*$ and write abstract groups multiplicatively, i.e $a*b \to ab$, $b*b^{-1} = bb^{-1}$
  - e, not 1, will still be identity

- <u>Thm 2.2</u> In a group, right & left cancellation hold, i.e $a*b = a*c \Rightarrow b = c$ & $ba = ca \Rightarrow b = c$
  <u>Proof</u> Suppose $ab = ac$. We group operation on left with a^{-1}. Thus $a^{-1}(ab) = a^{-1}(ac)$.
  Hence $(a^{-1}a)b = (a^{-1}a)c$, so $eb = ec$, and $b = c$. The proof for right cancellation is similar. $\square$
  (assoc. / inverses / identity)

  ★ don't just quote this thm. when using the result, actually apply the inverse

- <u>Thm. 2.3</u> Inverses are unique
  <u>Proof</u> Let $a \in G$ and suppose a^{-1} and b both act as inverses. Then $e = aa^{-1} = ab$.
  Group operation on left by an inverse of a, call it c, $c(aa^{-1}) = c(ab)$.
  Thus $(ca)a^{-1} = (ca)b$, hence $ea^{-1} = eb$, and $a^{-1} = b$. $\square$
  (assoc. / inverses / identity)

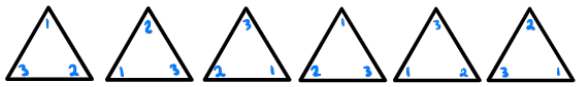
Dihedral Groups: The Example of D_3

Math 470: Abstract Algebra

Put the triangle like this △. For simplicity, make sure the vertices are labeled like the one on the board. Pick the triangle up, do some transformation, and put it back where it was. (Put it on the same "footprint" but with potentially different vertices in different places.)

1) How many different positions can the triangle end up in?

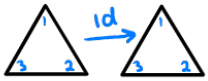
6 different positions



2) Shift from thinking about the positions to thinking about the transformations that get you from one position to another. We often call these transformations *symmetries*. We think of two transformations as being the same if they get the triangle from the same starting point to the same ending point. Come up with a scheme for labeling the transformations, and write all your transformations down. (Talk to the instructor before continuing on.)

We can:

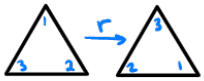
- do nothing (id)



- rotate again ($r_{240} = r \circ r$)

- rotate once (r)

120°



- flip (s)

3) The transformations are functions from the triangle to itself. We can define a binary operation on the transformations by composing them using function composition. Construct an operation table or Cayley table for the transformations. (Each row will be a labeled with a transformation element as will each column. The entry will be given by the composition of the row and column, with the row acting first.)

	id	r	s	r_{240}
id	$id \circ id$	$r \circ id$	$s \circ id$	$r_{240} \circ id$
r	$id \circ r$	$r \circ r$	$s \circ r$	$r_{240} \circ r$
s	$id \circ s$	$r \circ s$	$s \circ s$	$r_{240} \circ s$
r_{240}	$id \circ r_{240}$	$r \circ r_{240}$	$s \circ r_{240}$	$r_{240} \circ r_{240}$

"row acting first" puts row operation on the right

column ∘ row = column(row)

i.e $f \circ g = f(g)$

$f(x) \circ g(x) = f(g(x))$

4) Describe all the transformations in terms of just two. I like to use r for rotation clockwise by 120 degrees and s for the flip the fixes the top vertex.

$id = s \circ s$ $\quad$ $s = s$

$r = r$ $\quad$ $r_{240} = r \circ r$

☆ back side of WS not done in class